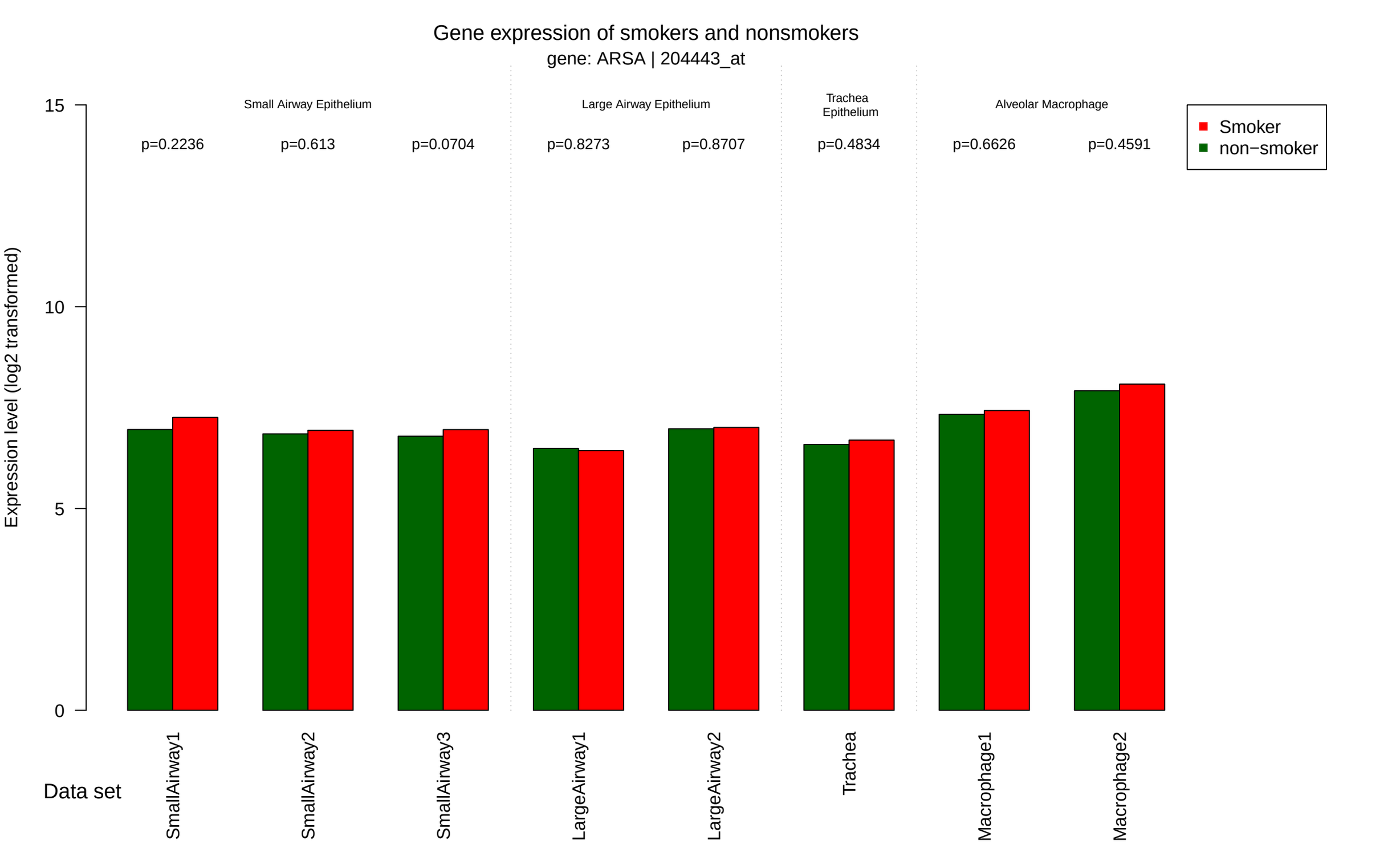
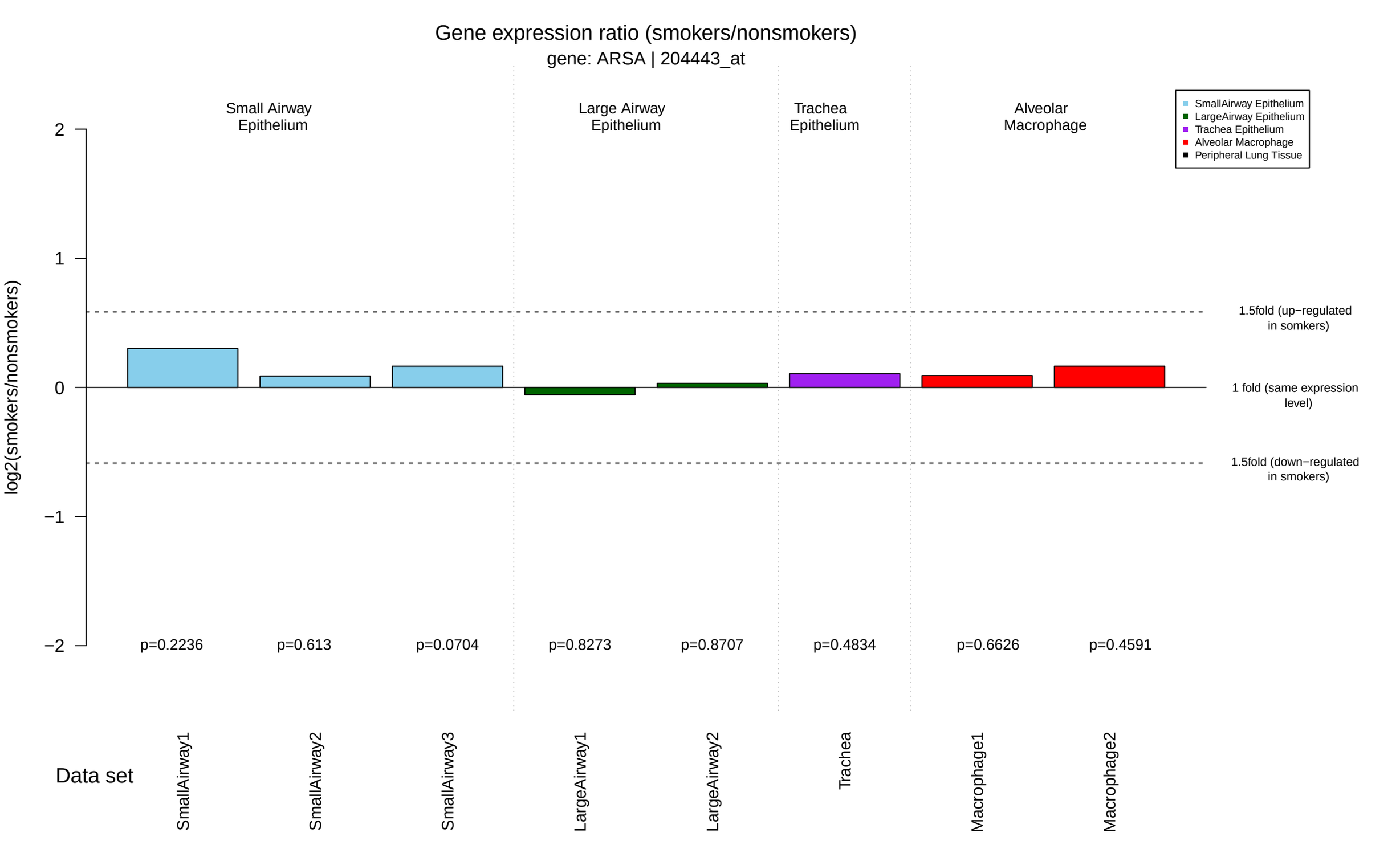
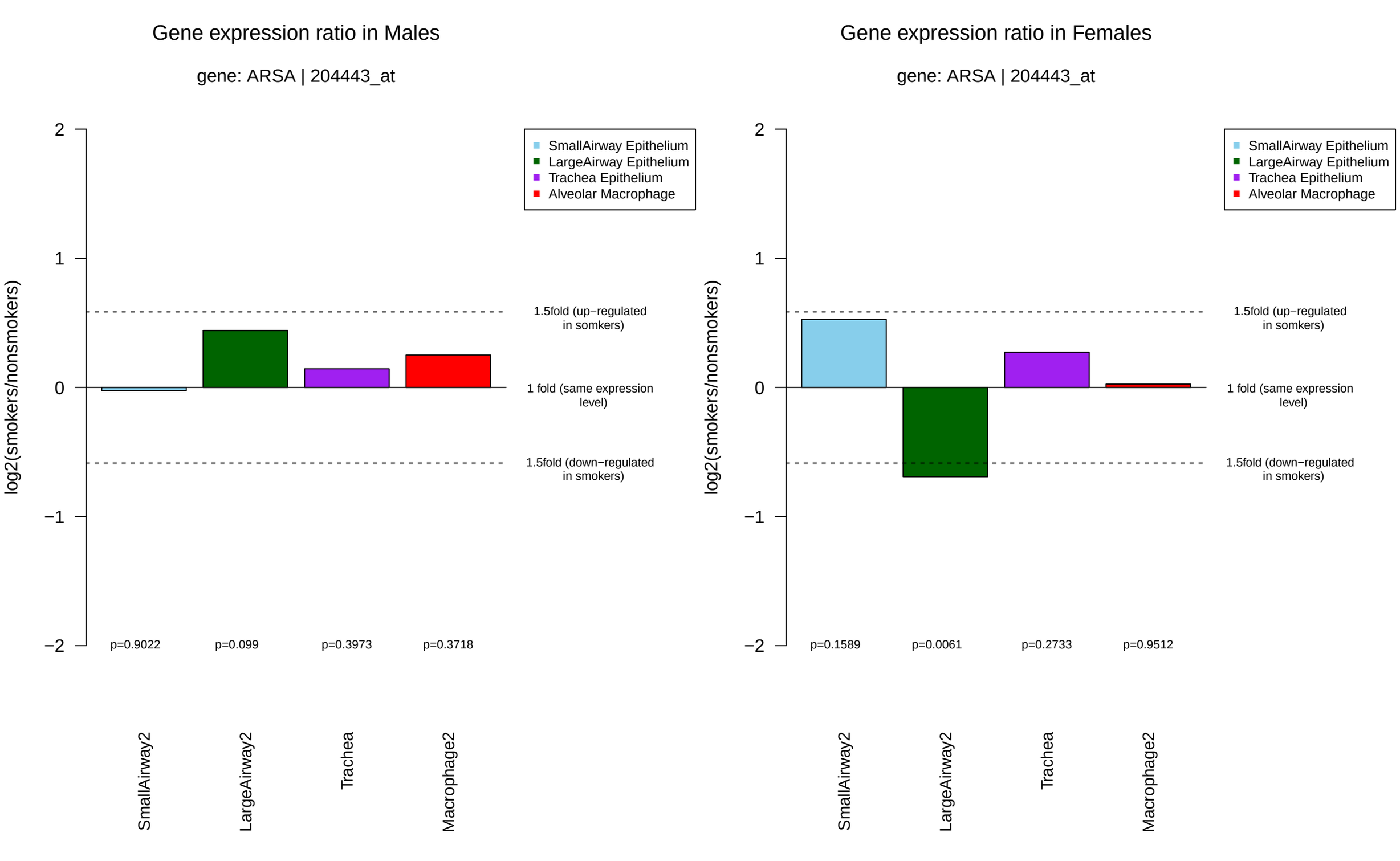
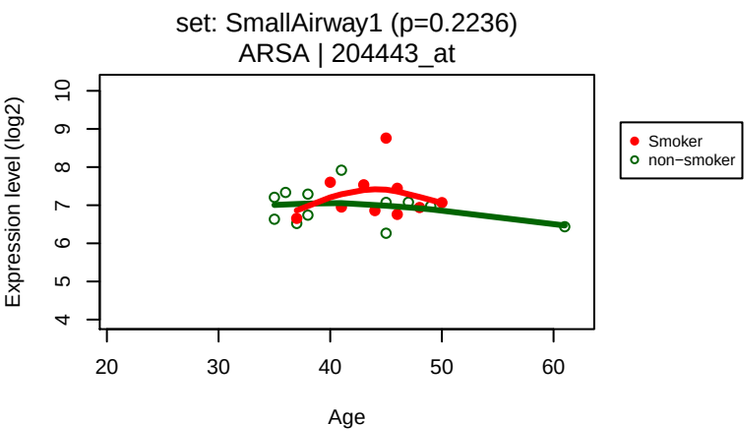
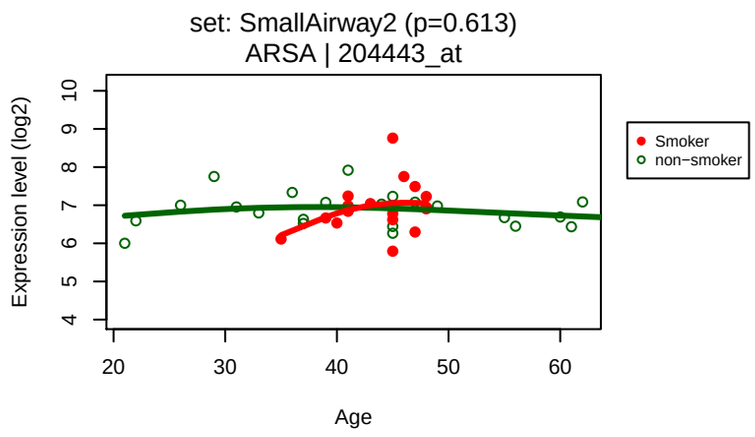
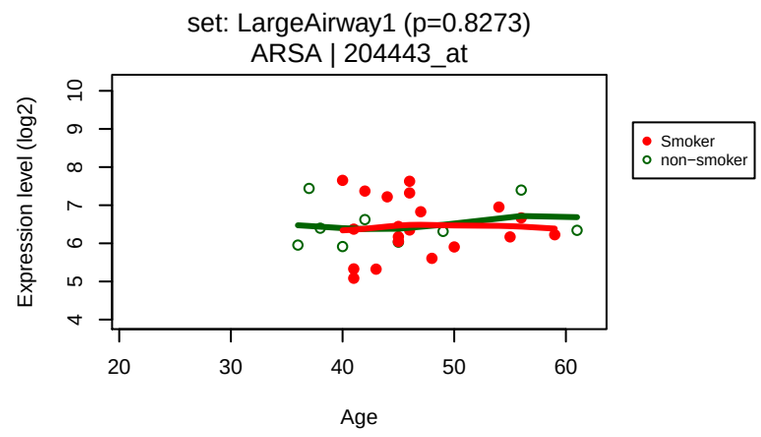

Gene expression of smokers and nonsmokers
gene: ARSA | 204443_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2236
p=0.613
p=0.0704
p=0.8273
p=0.8707
p=0.4834
p=0.6626
p=0.4591
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

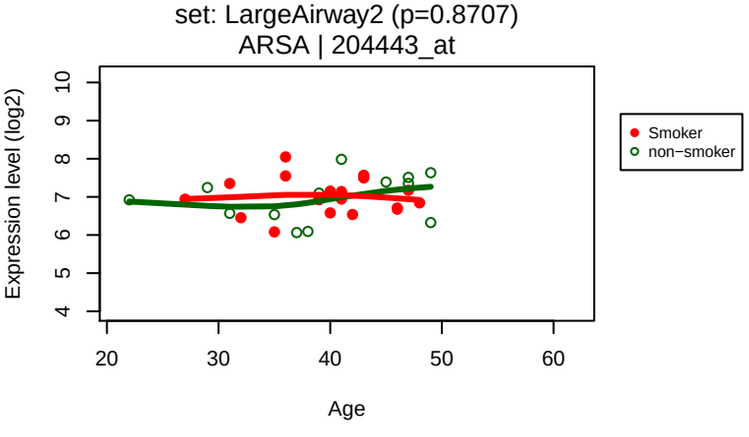

Gene expression ratio (smokers/nonsmokers)
gene: ARSA | 204443_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2236
p=0.613
p=0.0704
p=0.8273
p=0.8707
p=0.4834
p=0.6626
p=0.4591
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

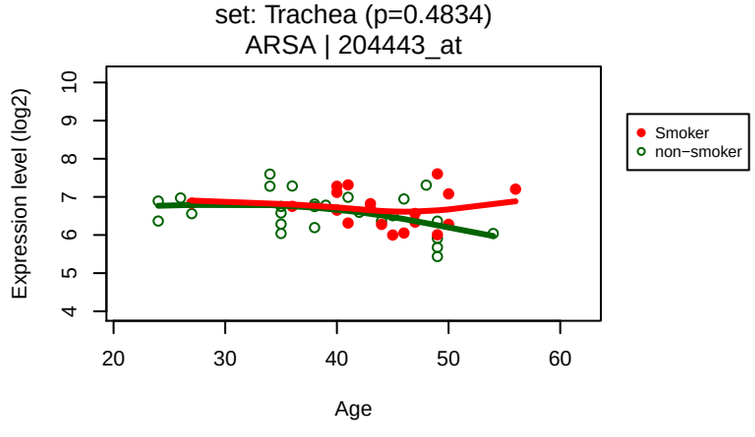

Gene expression ratio in Males
gene: ARSA | 204443_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9022
p=0.099
p=0.3973
p=0.3718
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARSA | 204443_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1589
p=0.0061
p=0.2733
p=0.9512
SmallAirway2
LargeAirway2
Trachea
Macrophage2

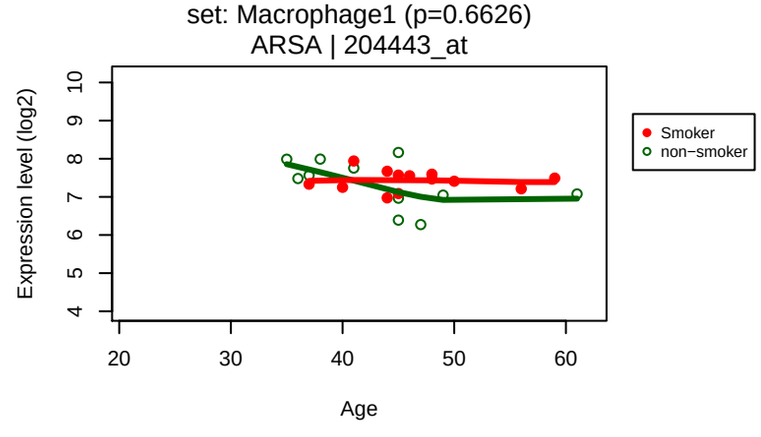

set: SmallAirway1 (p=0.2236)
ARSA | 204443_at
Expression level (log2)
Age
Smoker
non−smoker

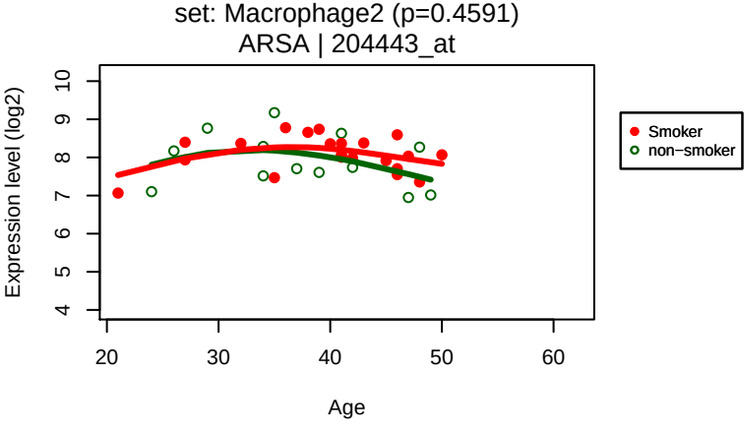

set: SmallAirway2 (p=0.613)
ARSA | 204443_at
Expression level (log2)
Age
Smoker
non−smoker

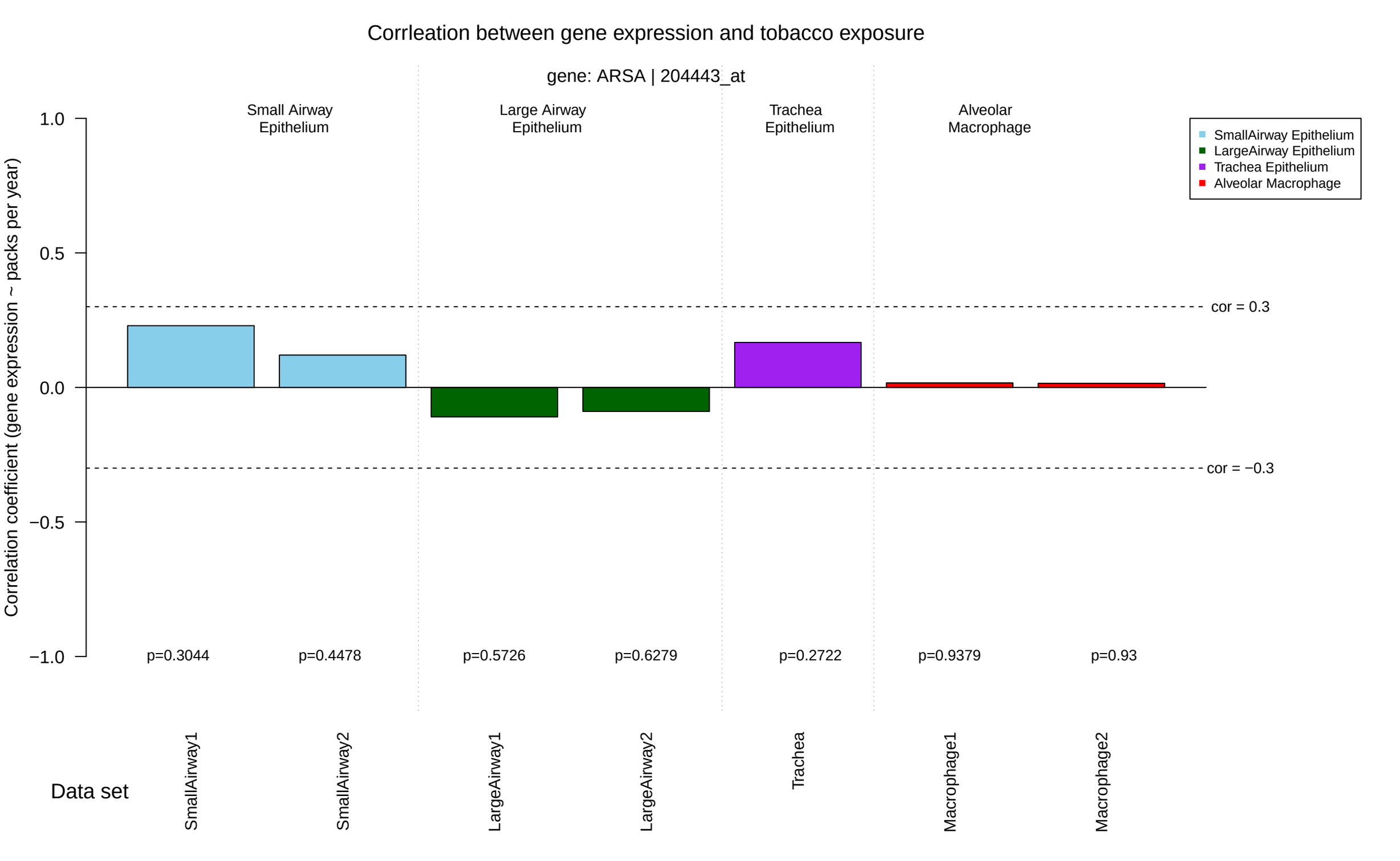

set: LargeAirway1 (p=0.8273)
ARSA | 204443_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8707)
ARSA | 204443_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4834)
ARSA | 204443_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6626)
ARSA | 204443_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4591)
ARSA | 204443_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARSA | 204443_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3044
p=0.4478
p=0.5726
p=0.6279
p=0.2722
p=0.9379
p=0.93
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2